

60870-5-101 data types, hot standby fail-over system, dual redundant real-time server and dual redundant monitor.

PHDL

PHDL is an HDL that functions as an alternative to mainstream graphical schematic capture tools. The language is compiled into a PCB netlist, which can then be imported into a layout tool. PHDL is currently on version 2.1. It includes an eclipse plugin version as well as a standalone command-line based version. Both function identically and output a netlist that can be imported into a PCB layout tool.

The PHDL compiler automatically supports the output of PADS and Eagle netlists. Through extending a simple Java class, users can generate a netlist in practically any format required by their choice of layout tool.

Papilio Loader

Papilio Loader is a tool used to load applications onto the Papilio Platform hardware. It is a set of Java files that work under Linux and Windows. These files simplify the interface to Open Source URJTAG application that does the task of loading application files to the Papilio Platform hardware. Papilio Loader is capable of loading Xilinx bitstream files.

ThSim

ThSim is a simulator for thermal conduction in solid material. It uses SPICE for calculations and wxWindows for providing Windows and Linux GUI. The solids analysed by this tool may consist of several block-shaped elements, which may have different material properties. A typical application is the simulation of peak temperatures in power semiconductor. The graphical output is either in line graph or in 3D using OpenGL display.

Analog Microcontroller Software Development Kit

The Analog Microcontroller Software Development Kit (AMSDK) provides a

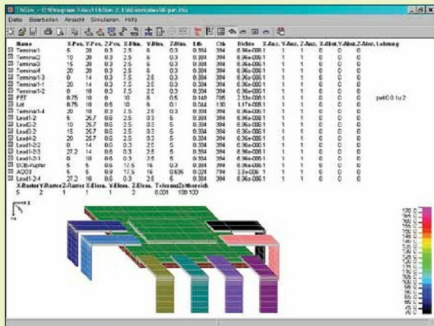


Fig. 3: Simulating thermal conduction in ThSim

high-level 'C' language programming environment and API supporting a quick and easy functional interface to Analog Microcontroller products. The API abstracts low-level details of various peripherals (controller registers, DMA engine and interrupt controller) into a simple, easy-to-use programming model.

The AMSDK is available in source form. Various example programs are also included. The kit supports the ADuCRF101 precision analogue microcontroller based on ARM Cortex M3 with ISM-band RF transceiver. The supported peripherals are common to all ARM Cortex-M3 Analog Microcontrollers.

The kit requires a third-party tool chain to compile, link, simulate, download, emulate, etc. Tool-chains such as IAR Embedded Workbench for ARM or Keil μ Vision4 MDK-ARM Microcontroller Development Kit are required for AMSDK.

MMANA

MMANA, also known as MMANA-GAL, is an antenna-analysing tool based on moment method, which was introduced in MININEC. The program provides the ability to change the language of signs and messages of the program.

Users can easily write a file in their own language by editing the English one. This means that MMANA can communicate with a user in any language, actually in the language of his operation system, and that users will be able to write language files themselves. Besides, users are now able to edit the program's signs, messages and help file in their own language.

NEC-2 is a useful utility for MMANA. It is intended for calculation of *.maa files and simulation of antenna models using input language NEC-2.

MMANA is also called MMANA-GAL, where GAL is a united abbreviation of names of DLIPBD and DL2KQ. This reflects substantial structural changes to the program. Besides regular improvements, new improvements create the possibility of further improvements of capabilities and services available with the program.

SimCAS

SimCAS is a free, Open Source, simple and very flexible analogue simulator. The goal of this project is not to build a simulator as powerful as Spice or Qucs but to try solving